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L19

PDF

Generate a lecture scribe intended to serve as exam reference material for a reading-based, closed-notes examination.

Use only the provided context, which consists of the lecture slides/PDF and the relevant textbook content.

The scribe must faithfully reconstruct what was taught in the lecture so that a student can rely on it later for revision.

Include definitions, notation, assumptions, theorem or result statements, proofs or derivations, and worked examples only if they appear in the provided context.

Do not introduce new examples, explanations, intuition, or material that is not present in the context.

Maintain strict faithfulness to the lecture's structure, scope, and logical flow.

Organize the scribe clearly with appropriate sections and headings so it can be used directly for exam preparation. make sure this is the only prompt no imporvisation would be done after this

Lecture 19: N-Variate Joint PDF/PMF/CDF

Course: CSE400 – Fundamentals of Probability in Computing

Date: March 17, 2026

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1. Joint Distribution of N Random Variables

The joint distribution describes the probability behavior of multiple random variables considered together.

- **Vector Representation:** For random variables $X_1, X_2, \ldots, X_N$, the joint behavior is expressed in vector form as:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- **Representation Methods:** Joint distributions can be represented using:
    - Joint Probability Mass Function (PMF)
    - Joint Probability Density Function (PDF)
    - Joint Cumulative Distribution Function (CDF)

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2. Jointly Gaussian Random Variables

The joint behavior of variables influenced by many random environmental processes can often be approximated by a Joint Gaussian Distribution.

2.1 Bivariate Gaussian PDF For two random variables X and Y, the joint PDF is given by:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y \quad 1-\rho^2} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)\right.\right.$$

- **Parameters:**
    - μ_X, μ_Y: Means of X and Y.
    - σ_X, σ_Y: Standard deviations of X and Y.
    - ρ: Pearson correlation coefficient.

2.2 Key Properties * **Marginal Distributions:** If X and Y are jointly Gaussian, their marginal distributions are strictly univariate Gaussians:
* $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$
* $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$

- **Marginal PDF Calculation:** Marginal PDFs are found by integrating the joint PDF:
    - $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy$
    - $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx$

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3. Worked Example: Jointly Gaussian RVs

Problem Statement: Two random variables X and Y are jointly Gaussian with $\mu_X = 2, \sigma_X^2 = 1, \mu_Y = -3, \sigma_Y^2 = 4$, and $\text{cov}(X, Y) = -1$.

(a) Find the Joint PDF $f_{XY}(x, y)$

1. **Find Standard Deviations:** $\sigma_X = \quad 1 = 1$ and $\sigma_Y = \quad 4 = 2$.
2. **Calculate Correlation Coefficient (ρ_{XY}):**

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1}{(1)(2)} = -0.5 \text{ [cite: 5]}$$

3. **Construct Final PDF:** Substituting $\mu_X = 2, \sigma_X = 1, \mu_Y = -3, \sigma_Y = 2, \rho_{XY} = -0.5$ into the general formula:

$$f_{XY}(x, y) = \frac{1}{2\pi \quad 3} \exp\left\{-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right\}$$

(b) Find Marginal PDFs $f_X(x)$ and $f_Y(y)$

Using the property that marginals are univariate Gaussians:

- $f_X(x) = \frac{1}{\quad 2\pi} \exp\left(-\frac{(x-2)^2}{2}\right)$
- $f_Y(y) = \frac{1}{2 \quad 2\pi} \exp\left(-\frac{(y+3)^2}{8}\right)$

(c) Find $\Pr(X < 0)$ in terms of the Q-function

1. **Identify CDF:** $\Pr(X < 0) = F_X(0)$.
2. **Standardize:** Using $Z = \frac{X - \mu_X}{\sigma_X}$ where $X \sim \mathcal{N}(2, 1)$:

$$\Pr(X < 0) = \Pr\left(\frac{X-2}{1} < \frac{0-2}{1}\right) = \Pr(Z < -2) = \Phi(-2) \text{ [cite: 3]}$$

3. **Apply Symmetry and Q-function Relation:**
    - Symmetry Property: $\Phi(-x) = 1 - \Phi(x) = \Pr(Z > x)$.
    - Q-function Definition: $Q(x) = \Pr(Z > x)$.
    - **Final Answer:** $\Pr(X < 0) = \Pr(Z > 2) = Q(2)$.

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4. Applications and Context

- **Climate Systems:** Used to study relationships such as Atmospheric CO_2 concentration vs. Global surface temperature anomaly.
- **Wireless Communication:** Application in Large MIMO Systems featuring M Transmit Antennas and N Receive Antennas (e.g., Massive MIMO in 5G and Giga-MIMO in 6G).